Chain Support Guides

# Bitcoin Support

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How to deposit and withdraw on Bitcoin from any Relay chain

Relay supports deposits to and withdrawals from Bitcoin across any supported chain. This guide covers the Bitcoin-specific parameters, PSBT signing, balance calculation, and API/SDK usage required for integration.

## Required Information

Bitcoin transactions differ from EVM chains in several ways. The sections below highlight the Bitcoin-specific parameters and requirements for your integration.

⚠ Bitcoin addresses are case-sensitive.

### SDK Properties

| Action | Parameter | Input | Description |
|---|---|---|---|
| Deposit to Bitcoin | `toChainId` | `8253038` | Chain ID assigned to Bitcoin in Relay. |
| | `recipient` | User's Bitcoin Address | Valid Bitcoin address. |
| | `toCurrency` | `bc1qqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqmql8k8` | Native BTC identifier. |
| Withdraw from Bitcoin | `chainId` | `8253038` | Chain ID assigned to Bitcoin in Relay. |

| Action | Parameter | Input | Description |
|---|---|---|---|
| | `currency` | `bc1qqqq

```
RELAY
async function getBitcoinBalance(address: string) {
    const response = await fetch(
        `https://mempool.space/api/address/${address}`
    );
    const data = await response.json();

    const fundedTxo = data.chain_stats.funded_txo_sum;
    const spentTxo = data.chain_stats.spent_txo_sum;
    const pendingSpent = data.mempool_stats.spent_txo_sum;

    // Balance in satoshis
    const balance =
        BigInt(fundedTxo) - BigInt(spentTxo) - BigInt(pendingSpent);

    return {
        balance, // Confirmed balance in satoshis
        pendingBalance: BigInt(pendingSpent), // Pending outgoing
    };
}
```

Balance values are returned in **satoshis** (1 BTC = 100,000,000 satoshis).

# API

## Params

| Action | Parameter | Input | Description |
|---|---|---|---|
| Deposit to Bitcoin | `recipient` | User's Bitcoin Address | Valid Bitcoin address. |
| | `destinationChainId` | `8253038` | Chain ID assigned to Bitcoin in Relay. |
| | `destinationCurrency` | `bc1qqqq

| Action | Parameter | Input | Description |
|---|---|---|---|
| RELAY | `originChainId` | `8253038` | Chain ID assigned to Bitcoin in Relay. |
| | `originCurrency` | `bc1qqqq

Request Response

```
curl -X POST "https://api.relay.link/quote/v2" \
  -H "Content-Type: application/json" \
  -d '{
    "user": "0x03508bb71268bba25ecacc8f620e01866650532c",
    "originChainId": 8453,
    "originCurrency": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
    "destinationChainId": 8253038,
    "destinationCurrency": "bc1qqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqmql8k8",
    "recipient": "bc1q4vxn43l44h30nkluqfxd9eckf45vr2awz38lwa",
    "tradeType": "EXACT_INPUT",
    "amount": "10000000"
  }'
```

Amounts are specified in the smallest unit of the currency. For BTC, this is **satoshis** (100000 satoshis = 0.001 BTC). For USDC, this is 6 decimal places (10000000 = 10 USDC).

## SDK

To use the SDK with Bitcoin, install and configure the . The adapter handles PSBT signing and transaction broadcasting.

```
npm install @relayprotocol/relay-bitcoin-wallet-adapter
```

For implementation details and code samples, see the .

## Deposit Addresses (Optional)

Relay also supports the , which allows bridging without wallet connection or PSBT signing. When using deposit addresses, there are additional considerations:

## Block Confirmations

When using deposit addresses, Relay waits for **1 block confirmation** before processing. Bitcoin's block time averages ~10 minutes but varies significantly. blocks can arrive within minutes of each other or take 30-50 minutes.

> ⚠ When polling for deposit address transactions, allow for at least 2-3 blocks worth of time before considering a transaction stalled. You can monitor block times at **mempool.space**.

## Quote Requirements for Deposit Addresses

When using `useDepositAddress: true` with Bitcoin as origin:

- The `user` address must be a valid Bitcoin address
- The `user` address balance must exceed the quoted amount (otherwise returns `INSUFFICIENT_FUNDS`)
- For indicative quotes, use a known high-balance address ("whale address"). The actual deposit can originate from any address.

Request Response

```
curl -X POST "https://api.relay.link/quote/v2" \
  -H "Content-Type: application/json" \
  -d '{
    "user": "bc1q4vxn43l44h30nkluqfxd9eckf45vr2awz38lwa",
    "originChainId": 8253038,
    "originCurrency": "bc1qqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqmql8k8",
    "destinationChainId": 8453,
    "destinationCurrency": "0x833589fcd6edb6e08f4c7c32d4f71b54bda02913",
    "recipient": "0x03508bb71268bba25ecacc8f620e01866650532c",
    "tradeType": "EXACT_INPUT",
    "amount": "100000",
    "useDepositAddress": true,
    "refundTo": "bc1q4vxn43l44h30nkluqfxd9eckf45vr2awz38lwa"
  }'
```

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